Consequences of Treating Reactions as First-Class Objects

Robust to annotation gaps

PMDs are computed for any peak pair — annotated or not. Missing IDs no longer block biological inference.

Chemically interpretable

Δm = 2.0157 → reduction
Δm = 14.0157 → methylation
The network is readable without a lookup table.

Comparable across systems

Reaction networks can be compared across conditions, species, and sample types in ways metabolite lists cannot.

Links to pathway knowledge

Connects to KEGG / HMDB / Reactome when IDs exist, yet remains usable when annotation is incomplete.